

COVE: COntext and VEracity prediction for out-of-context images

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Abstract

Images taken out of their context are the most prevalent form of multimodal misinformation. Debunking them requires (1) providing the true context of the image and (2) checking the veracity of the image's caption. However, existing automated fact-checking methods fail to tackle both objectives explicitly. In this work, we introduce **COVE**, a new method that predicts first the true **CO**ntext of the image and then uses it to predict the **VE**racity of the caption. COVE beats the SOTA context prediction model on all context items, often by more than five percentage points. It is competitive with the best veracity prediction models on synthetic data and outperforms them on real-world data, showing that it is beneficial to combine the two tasks sequentially. Finally, we conduct a human study that reveals that the predicted context is a reusable and interpretable artifact to verify new out-of-context captions for the same image. Our code and data are made available.¹

1 Introduction

An out-of-context (OOC) image is a form of multimodal misinformation where the caption misrepresents the image in one or several dimensions, including the date, location, or event depicted (Luo et al., 2021; Dufour et al., 2024). In 2023, more than 40% of the visual misinformation verified by fact-checkers consisted of OOC images (Dufour et al., 2024). To debunk them, human fact-checkers follow two objectives: identifying the true context of the image, which usually takes the form of a fixed set of items (Silverman, 2013; Urbani, 2020; Mossou and Higgins, 2021; Khan et al., 2023, 2024), and deciding on the veracity of the caption. Identifying the true context of the image first is often beneficial when checking the caption, as it may reveal inconsistencies with the image. However, it goes beyond that objective, as most context

¹github.com/UKPLab/naacl2025-cove

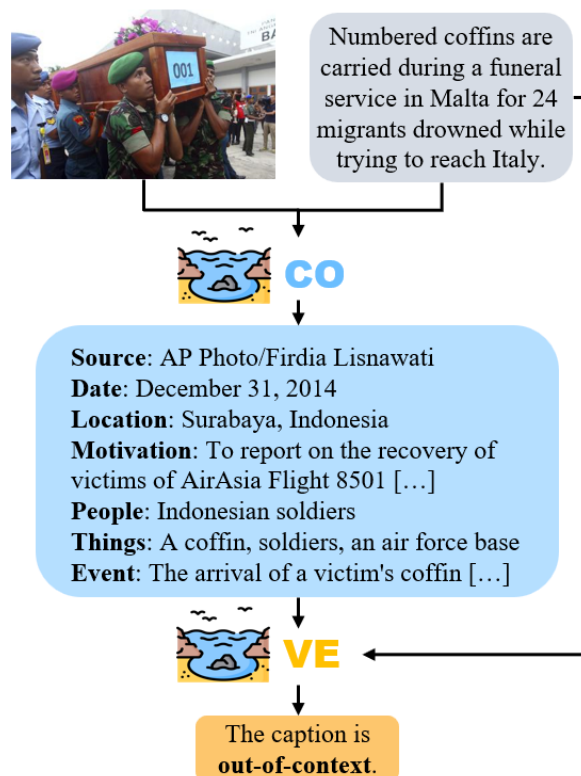


Figure 1: The two steps of **COVE**: (1) Generating the true context of the image. (2) Predicting the veracity of a caption by comparing it with the generated context.

items are neither consistent nor inconsistent with the caption and provide more details than needed to verify the caption. In Figure 1, the context, shown in blue, reveals that the *location* in the caption is inconsistent: Malta instead of Indonesia. It also discusses the *source* and *date*, which are absent from the caption, and provides a precise *location* at the city level.

Many methods have been proposed to facilitate fact-checking (FC) of OOC images. However, they study either context prediction (Tonglet et al., 2024) or veracity prediction (Abdelnabi et al., 2022; Papadopoulos et al., 2023, 2024a; Qi et al., 2024). They do not consider automating them sequentially,

leveraging the comprehensive context to predict the veracity of the caption.

In this work, we introduce **COVE**, the first method that both predicts a comprehensive **Context** for the image and a **Veracity** label for the caption. The method consists of two steps, shown in Figure 1. First, the context is predicted as a set of seven items, three of which are first introduced in this work. Compared to prior work, COVE leverages a more diverse set of evidence to predict the context, including web search, knowledge bases, and the parametric knowledge of large language models (LLMs). Afterward, the caption is compared with the context to predict its veracity. We summarize our contributions as follows. (1) Leveraging a more diverse set of evidence, COVE outperforms the context prediction SOTA (Tonglet et al., 2024) for all context items, including three items introduced in this work, by 0.3 to 18.9 percentage points. (2) COVE is competitive with the best models for veracity prediction on synthetic data and outperforms them on real-world data by up to 4.5 percentage points in Macro F1, highlighting the benefits of automating veracity prediction based on the predicted context. (3) Our experiments show that the predicted context is an interpretable artifact for human users, which they can reuse multiple times to verify new captions about the same image.

2 Related work

Veracity prediction for OOC images

Providing a veracity label for OOC images has received significant attention in automated fact-checking (AFC) research (Akhtar et al., 2023). It is a binary classification task with labels {accurate, OOC}. Synthetic datasets have been created by replacing entities in the true caption (Sabir et al., 2018; Müller-Budack et al., 2020) or by mismatching image-caption pairs from news corpora (Luo et al., 2021). Smaller datasets based on real-world fact-checks have recently been proposed (Aneja et al., 2023; Papadopoulos et al., 2024b; Pham et al., 2024). Models that leverage external evidence outperform those that use only the image and the caption as input (Luo et al., 2021; Zhang et al., 2023b). Abdelnabi et al. (2022) collects text evidence with reverse image search and image evidence by querying a search engine with the caption. The performance can further be improved by predicting the stance of the evidence towards the caption (Yuan et al., 2023) or ranking

the most relevant evidence (Papadopoulos et al., 2023). Unlike prior methods, ECENet provides an explanation in the form of a summary of the most relevant text evidence (Zhang et al., 2023a). Recent work leverages multimodal LLMs (MLLMs) with instruction-tuning (Qi et al., 2024), external tools (Braun et al., 2024), or multi-agent debate (Lakara et al., 2024). Papadopoulos et al. (2024a) showed that simple classifiers like random forest trained on top of the image, caption, and image and text evidence embeddings achieve SOTA performance. Their results show that veracity can often be predicted based on shallow heuristics, highlighting the need to assess the progress in the field from other perspectives (Papadopoulos et al., 2024a), such as context prediction.

Context prediction for OOC images

Context prediction, or image contextualization, has received less attention in AFC. Tonglet et al. (2024) formulate it as a question-answering (QA) task where each context item is predicted given the corresponding question and a set of evidence which may include the image, the caption, or pieces of information derived from them. Tahmasebi et al. (2025) formulate it as a true/false classification task where candidate *people*, *locations*, and *events* are verified with a MLLM. 5Pils is a real-world dataset (Tonglet et al., 2024) which contains context labels based on human fact-checking practices (Urbani, 2020; Mossou and Higgins, 2021; Khan et al., 2023, 2024). Tonglet et al. (2024) proposed a baseline for context prediction that retrieves text evidence with reverse image search and answers context questions with a (M)LLM and the image and the evidence as input.

In this work, we fill an important gap by introducing the first method that performs both tasks sequentially, leveraging the generated context for veracity prediction.

3 Context and Veracity (COVE)

COVE is a method to predict the true context of an image first and then the veracity of its caption. The overall architecture can be divided into six steps, as illustrated in Figure 2. The first three steps are concerned with the collection of a diverse set of evidence, i.e., web captions and visual entities (§3.1), Wikipedia entities (§3.2), and automated captions (§3.3). Based on their similarity with web images

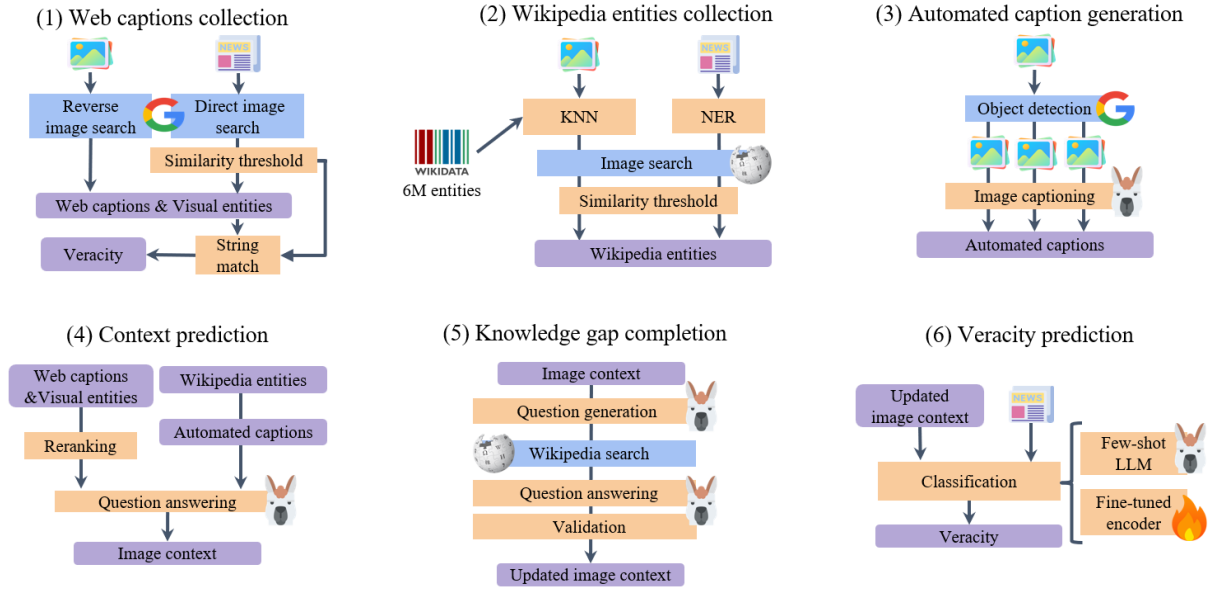


Figure 2: The architecture of COVE consists of six steps. The first three are performed in parallel and consist of retrieving evidence. Step 4 predicts the context items in a QA setting. Step 5 updates missing items based on the existing ones and Wikipedia knowledge. Step 6 predicts the veracity of the caption based on the predicted context.

and captions, the veracity of certain instances is already predicted during evidence retrieval (§3.1). After collecting the evidence, a first version of the context is predicted with an LLM (§3.4). Then, some missing context items are updated by searching relevant Wikipedia passages (§3.5). Eventually, a model predicts the veracity of the caption based on the predicted context (§3.6).

3.1 Web captions collection

Following Abdelnabi et al. (2022), we use reverse image search with the Google Vision API to retrieve web captions and visual entities associated with the same or partially matching images,² and a custom Google search engine to retrieve relevant web images given the caption.³ We compute the cosine similarity between the CLIP (Radford et al., 2021) embeddings of the instance’s image and the web images. For images with similarity above a threshold t_{match} , their attached caption is added to the set of web captions.

The web captions are sufficient to assign a veracity label to some instances, especially those for which the caption and/or the image come directly from newspapers. We apply the following three rules, which we refer to as veracity rules: (1) if the caption is an exact string match with one or more reverse image search web captions, the ve-

racity is accurate, (2) if the similarity with a web image is above t_{match} , the veracity is accurate, (3) if the similarity with a web image is below threshold t_{non_match} and the attached web caption is an exact string match with the caption to verify, the veracity is OOC.

3.2 Wikipedia entities collection

Images often contain entities, such as celebrities, products, and landmarks, that can be paired with a Wikipedia page (Müller-Budack et al., 2020). Recognizing these entities and providing them as evidence can help predict the context. An example is illustrated in Figure 3. First, we collect a set of candidate entities: (1) we collect named entities from the caption and normalize them to match one or more Wikipedia entries using GENRE (Cao et al., 2021), (2) we use the OVEN index (Hu et al., 2023a), which contains 6 Million Wikidata entities, and retrieve the k nearest neighbors based on the cosine similarity between their CLIP text embeddings and the image embedding. For each candidate entity, if its embedding similarity with the image is superior to threshold t_{wiki_text} , it is retained. Otherwise, we scrape up to three images from the corresponding Wikipedia page and compute their embedding’s cosine similarity with the image. If at least one similarity score is higher than t_{wiki_image} , the entity is retained.

²cloud.google.com/vision/docs/detecting-web

³programmablesearchengine.google.com/about/

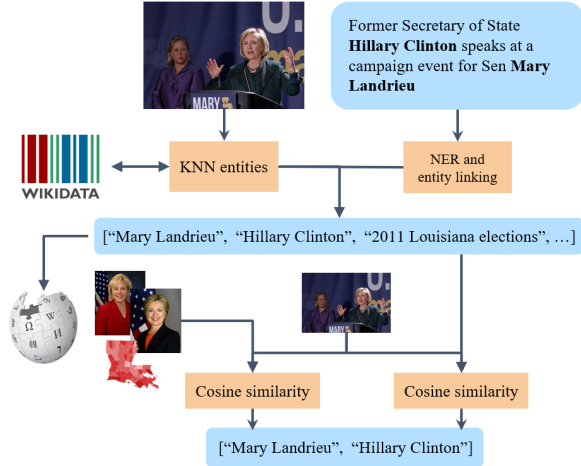


Figure 3: Wikipedia entities collection. The candidate set is composed of the entities in the caption and those that are most similar to the image. Candidates are retained if the similarity between the image and their name or their Wikipedia images passes a threshold.

3.3 Automated caption generation

The third set of evidence is generated automatically with the MLLM LlavaNext (Liu et al., 2023, 2024). One caption describing the entire image and another one describing the people are generated by providing the entire image as input. Then, following Hu et al. (2023b), we detect objects in the image using the Google Vision API.⁴ We crop the image to its bounding boxes, and caption it with LlavaNext using a question specific to the object’s category. Categories are manually defined with their questions in Appendix A.

3.4 Context prediction

We predict seven context items: $\{source, date, location, motivation, people, things, event\}$. The first four items were introduced in Tonglet et al. (2024), while the last ones are new to this work. The first l web captions are ranked based on the number of named entities relevant to the context item that they contain, for example, the number of PERSON entities for the *people* item. The selected Wikipedia entities and the automated captions are merged into two captions: one for everything that relates to *people* in the image and the other one for all other pieces of information. All the collected evidence is provided as input with a question to the LLM Llama 3 (MetaAI, 2024) to predict a context item. When the evidence does not contain the answer, the model is prompted to output “Unknown”. Each

⁴cloud.google.com/vision/docs/object-localizer

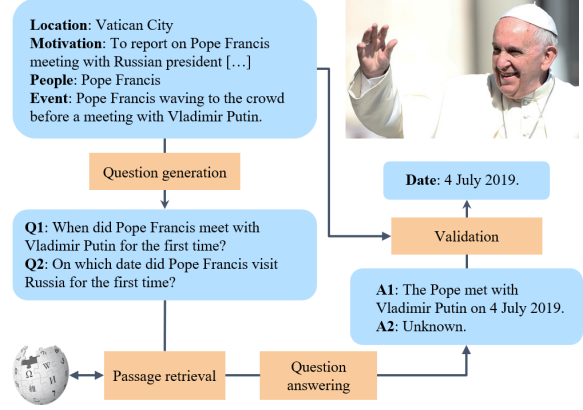


Figure 4: Knowledge gap completion. Questions are generated based on the predicted context and answered with Wikipedia passages. If the answers are relevant, the context is updated.

item, its associated question, and relevant named entities are discussed in Appendix B .

3.5 Knowledge gap completion

The *date* and *location* can often be missing after step 4. However, these two items can be derived from other context items and world knowledge. Following QACheck (Pan et al., 2023a), we use a sequence of three modules: question generation, question answering, and validation, illustrated in Figure 4. If the answer to *date* is Unknown, and *location* + *people*, *event*, or *motivation* are available, we provide the currently known context items as input to Llama 3 and ask it to generate up to three questions that could help predict the date of the image. For each question, relevant Wikipedia passages are retrieved with WikiChat and Colbert (Khattab and Zaharia, 2020; Semnani et al., 2023) and provided as input to Llama 3 to provide an answer. Finally, Llama 3 predicts the *date*, if it can be determined, based on the existing context and the generated QA pairs. For *location*, we apply the same pipeline if the *date* + *people*, *event*, or *motivation* are available. More examples of knowledge gap completions are shown in Appendix C.

3.6 Veracity prediction

This step takes the predicted context and the caption as input and is performed only for captions that could not automatically be verified based on the web evidence (§3.1). We consider two different models. (1) A frozen Llama 3 with few-shot demonstrations as input. Each demonstration’s output starts with an explanation of the inconsistencies, followed by the predicted veracity, which is

either “accurate”, “OOC”, “Unknown, probably [label]”, which counts as a prediction of the corresponding label, or “Unknown”, which is mapped to the majority predicted label. This flexibility in the output answers was empirically found to help Llama 3 in ambiguous cases. (2) A DebertaV3 (He et al., 2023) model fine-tuned on (predicted context, ground truth veracity) training pairs.

4 Experiments

4.1 Datasets

NewsCLIPpings is a synthetic dataset for veracity prediction (Luo et al., 2021). Images and their captions are selected from Visual News (Liu et al., 2021). OOC pairs are created by mismatching images and captions using various measures of semantic similarity. Following prior works, we use the “merged-balanced” split, which contains 71,072, 7,024, and 7,264 instances in the train, validation, and test splits, respectively. Within a split, each caption appears twice, once as accurate and once as OOC. NewsCLIPpings does not contain ground truth context items. Hence, we create them by decomposing the accurate caption in a set of context items with Llama 3, as explained in Appendix D.

5Pils-OOC is a real-world test set containing 624 images, each paired with two captions, one accurate and the other OOC, for a total of 1248 instances. We construct 5Pils-OOC as a subset of 5Pils (Tonglet et al., 2024). The images and OOC captions have been fact-checked by human experts from three organizations: Factly, Pesacheck, and 211Check. The images come from both Western and non-Western contexts, in particular, India and Ethiopia. As 5Pils does not contain accurate captions, we generate them automatically using GPT4 (OpenAI, 2023) based on the ground truth context items. Furthermore, 5Pils does not contain ground truth labels for context items *people*, *things*, and *event*. They are derived from the accurate caption, as explained in Appendix D. The creation of 5Pils-OOC from 5Pils is detailed in Appendix E.

4.2 Context metrics

We report one metric per context item. Additional evaluations with all metrics introduced in Tonglet et al. (2024) are provided in Appendix F.

Source, *motivation*, *things*, and *event* are evaluated with Meteor (Banerjee and Lavie, 2005).

Date predictions are mapped to timestamps. We use Δ , which is inversely proportional to the dis-

tance in years with the ground truth.

Location predictions are mapped to coordinates using GeoNames.⁵ We use Coordinates Δ (CO Δ), which is inversely proportional to the distance in thousand kilometers with the ground truth.

People expects sets of named entities as predictions, evaluated with the Macro F1-score (F1).

4.3 Veracity metrics

We use four metrics for veracity prediction: the accuracy A, the recall over accurate samples R_{ACC} , the recall over OOC samples R_{OOC} , and F1.

4.4 Baselines

For context prediction, we compare COVE with the 5Pils baseline Tonglet et al. (2024), which provides the image and web captions from reverse image search as input to a MLLM, LlavaNext in our case, asking one question per context item.

We compare COVE against three SOTA models for veracity prediction of OOC images, which all rely on external evidence. **RED-DOT** (Papadopoulos et al., 2023) and **AITR** (Papadopoulos et al., 2024a) use transformer architectures trained on top of the CLIP embeddings of the image, the caption, and text and image evidence. Furthermore, they filter the evidence to keep only the most relevant one from each modality. **SNIFFER** (Qi et al., 2024) predicts veracity based on two signals: (1) the detection of inconsistencies between the caption, the image, and the visual entities, using a fine-tuned InstructBLIP (Dai et al., 2023), and (2) the detection of inconsistencies between the caption and text evidence using a frozen Vicuna (Chiang et al., 2023), which we replace here by Llama 3 for a fair comparison. We also evaluate veracity prediction with Llama 3 based on the context items predicted by the 5Pils baseline. Finally, we report as an upper bound the COVE veracity results using the ground truth context items.

4.5 Implementation details

All hyperparameters are tuned, and ablations are performed on a random sample of 1500 instances from the NewsCLIPpings validation set. DebertaV3 is fine-tuned on a subset of the NewsCLIPpings train set containing 5000 instances. Inference with Llama 3 is done in a few-shot setting with 4 to 8 demonstrations, which have been selected and labeled by hand from a random sample of 100

⁵geonames.org

	Source (M)	Date (Δ)	Loc. (CO Δ)	Mot. (M)	People (F1)	Things (M)	Event (M)
NewsCLIPpings							
5Pils baseline	3.4	28.9	32.7	1.5	39.9	9.9	15.4
COVE	8.1	41.5	51.6	11.6	49.0	10.9	22.4
5Pils-OOC							
5Pils baseline	0.3	1.8	21.8	3.0	12.8	4.9	4.8
COVE	0.6	7.0	28.9	15.1	20.5	7.2	9.4

Table 1: Context prediction results on the test sets (%). The best scores are marked in **bold**. Loc. and Mot. are the location and the motivation, respectively.

	NewsCLIPpings				5Pils-OOC			
	(A)	(R _{ACC})	(R _{OOO})	(F1)	(A)	(R _{ACC})	(R _{OOO})	(F1)
RED-DOT 🔴	90.3	87.3	93.3	90.3	46.8	42.8	50.8	46.7
AITR 🔴	93.5	94.8	92.1	93.5	52.6	81.4	23.9	48.4
SNIFFER 🔴❄️	88.4	92.3	84.4	88.3	56.3	86.4	26.1	51.9
5Pils baseline - Llama 3 ❄️	77.2	74.6	79.8	77.2	55.7	42.8	68.6	54.9
COVE with predicted context								
DebertaV3 🔴	87.9	84.0	91.8	87.9	56.7	47.6	65.9	56.4
Llama 3 ❄️	86.7	83.5	89.9	86.7	58.2	33.0	83.3	55.3
COVE with ground truth context								
DebertaV3 🔴	94.1	99.2	88.9	94.0	80.7	100.0	61.4	79.9
Llama 3 ❄️	95.1	97.1	93.1	94.4	95.9	99.7	92.1	95.3

Table 2: Veracity prediction results on the test sets (%). 🔴 and ❄️ indicate trained models and frozen models used in a few-shot setting, respectively. The best scores without ground truth are marked in **bold**.

NewsCLIPpings train instances. Model versions and the hyperparameters are listed in Appendix G. Appendix H reports the prompts for Llama 3.

The same set of web evidence is used for all methods. Following RED-DOT, AITR, and SNIFFER, we only use the caption field of the web evidence for NewsCLIPpings. For 5Pils-OOC, we use the title field because few web evidence provides a caption. This is more restrictive than the setup of Tonglet et al. (2024) and limits context prediction by ignoring important webpage fields like the publication date. However, it ensures a fair comparison with the veracity prediction baselines. Unlike NewsCLIPpings, the web evidence of 5Pils-OOC are multilingual, including texts in Amharic, Arabic, Hindi, and Telugu.

4.6 Main results

Context prediction

Thanks to its more diverse evidence set, COVE outperforms the 5Pils baseline (Tonglet et al., 2024) on all context items and both datasets, as shown in Table 1. The performance increases from 1.0 to 18.9 on NewsCLIPpings, and from 0.3 to 12.1 percentage points on 5Pils-OOC. For *date*, *location*, *motivation*, and *event*, COVE is not only more accurate but also abstains less often from answering than the baseline. We attribute these improvements to the larger and more diverse evidence set and the knowledge gap completion step for *date* and

location. In particular, on 5Pils-OOC, including the knowledge gap completion more than doubles the COVE *date* scores, from 3.3 to 7.0%, and increases the *location* scores by 6.1 percentage points. Thanks to the Wikipedia entities retrieval, COVE achieves up to 9.1 percentage points higher F1 than the baseline for *people*. *Source* relies a lot on web captions, which are also part of the evidence set in the baseline, resulting in limited improvements.

With both methods, better contextualization is achieved on NewsCLIPpings. The largest drops in performance when moving to 5Pils-OOC are observed for *source*, *date*, *location*, and *people*. The following properties of the images in NewsCLIPpings explain this: (1) they are of higher quality, (2) they all originate from news articles and the retrieved web captions contain more information and are less noisy, impacting *source* the most, (3) most of them are set in a Western context, on which MLLMs tend to perform better (Ananthram et al., 2024), while most of 5Pils-OOC images are set in East Africa and South Asia. This affects *location* the most, which is often estimated based on the automated captions if no web captions are available.

Veracity prediction

The veracity results of COVE are competitive with the baselines on NewsCLIPpings, as shown in Table 2, suffering mainly from a low R_{ACC} while achieving near SOTA R_{OOO}. COVE with DebertaV3 achieves slightly better performance, which we attribute to the fine-tuning on NewsCLIPpings train instances.

All methods suffer from lower results when switching from synthetic data to the 5Pils-OOC real-world data. However, COVE becomes the strongest method, achieving better accuracy, R_{OOO}, and F1 than the baselines. While COVE with Llama 3 achieves the best accuracy, COVE with DebertaV3 is the best method in terms of F1. In comparison, RED-DOT performs worse than random, and AITR and SNIFFER have very low R_{OOO}, as low as one OOC caption on four. RED-DOT and AITR always predict the OOC label for instances that have web captions, highlighting their reliance on shallow heuristics that do not generalize beyond NewsCLIPpings (Papadopoulos et al., 2024a). While fine-tuned on synthetic data like the baselines, DebertaV3 generalizes better to 5Pils-OOC, which we attribute to leveraging a comprehensive and structured context as input.

Web captions	Visual entities	Wikipedia entities	Automated captions	Knowledge gap	Context prediction	Veracity prediction	Veracity rules	Source (M)	Date (Δ)	Location (CO Δ)	Motivation (M)	People (F1)	Things (M)	Event (M)	Veracity			
															(A)	(R _{ACC})	(R _{OOO})	(F1)
✓	✓	✓	✓	✓	✓	✓	✓	11.2	48.2	55.7	11.2	53.0	10.0	22.2	88.3	86.8	89.8	88.3
✓	✗	✗	✗	✗	✗	✗	✗	-	-	-	-	-	-	-	53.1	69.8	35.5	44.4
✓	✓	✓	✓	✓	✓	✓	✓	11.2	48.2	55.7	11.2	53.0	10.0	22.2	84.2	80.2	88.4	84.2
✓	✓	✓	✓	✓	✗	✓	✓	-	-	-	-	-	-	-	78.2	94.9	60.7	77.4
✓	✓	✗	✗	✗	✓	✓	✓	11.2	43.7	49.6	10.8	45.1	11.1	17.7	85.1	94.1	75.7	78.6
✓	✗	✓	✓	✗	✓	✓	✗	14.1	48.2	54.5	10.7	53.6	10.7	21.0	88.3	88.5	88.1	88.3
✗	✓	✓	✓	✗	✓	✓	✓	1.5	9.6	14.8	9.3	28.1	6.9	9.5	77.9	73.2	82.8	77.9
✗	✗	✓	✓	✗	✓	✓	✗	1.5	8.2	10.1	9.1	27.5	7.4	8.5	77.1	73.0	81.4	77.1
✗	✗	✓	✓	✓	✓	✓	✗	1.5	8.8	11.8	9.1	27.5	7.4	8.5	76.8	71.1	82.8	76.8

Table 3: Ablation results on the NewsCLippings validation subset (%). Veracity metrics are reported for few-shot Llama 3. ✓ and ✗ indicate included and removed components, respectively. The best scores are marked in **bold**.

Providing the context items predicted by the 5Pils baseline as input to Llama 3 results in a drop in performance, in particular on NewsCLippings where the difference in context prediction performance between the 5Pils baseline and COVE is the largest. Furthermore, we validate that the COVE results using the ground truth context lead to near-perfect R_{ACC}, both with DebertaV3 and Llama 3. This is expected because the ground truth context items are obtained by decomposing accurate captions. On NewsCLippings, COVE achieves F1 of 94.4 and 94.0 % with Llama 3 and DebertaV3, respectively, higher than all baselines. These results confirm that predicting a comprehensive and accurate context and providing it as input ensures high performance for veracity prediction. While Llama 3 with ground truth context achieves a similar performance on 5Pils-OOC, DebertaV3 faces a large drop in R_{OOO}. We assume that DebertaV3 suffers like the baselines, although to a lower extent, from being trained on synthetic OOC data, which follow a different distribution than the real-world data of 5Pils-OOC.

4.7 Ablation study

We report ablation results on the validation subset in Table 3, using Llama 3 for veracity prediction. By relying solely on the veracity rules for web captions (§3.1), an accuracy of 53.1% is obtained, and more than a third of the OOC captions can be detected. This happens without predicting any context element.

Removing veracity rules for web captions deteriorates R_{ACC}, and R_{OOO} to a lower extent. Removing context prediction, providing instead the raw

evidence as input for veracity prediction, decreases R_{OOO} by 29.1 percentage points. This shows that using the predicted context as input is a decisive factor favoring R_{OOO} over R_{ACC}.

The accuracy decreases, and a large imbalance in favor of R_{ACC} appears when using only web captions and visual entities. Furthermore, the performance on several context items decreases too, by up to 7.9 percentage points for *people*, highlighting the need to consider a more diverse set of evidence. For *source* and *people*, the best results are achieved without visual entities, indicating that they may contain irrelevant information and conflict with the other evidence. All context and veracity metrics experience a large drop when removing web captions. In particular, *source* relies the most on web captions. Removing both the visual entities and web captions decreases performance further. However, the F1 for *people* remains more than half of what would be obtained with web results, thanks to the Wikipedia entities collection step. Furthermore, COVE still detects more than 80% of the OOC captions, and the F1 for veracity is only 1.5 percentage points lower than the one obtained with web captions and visual entities only. In the absence of web results, knowledge gap completion can improve *date* and *location* scores but only to a small extent.

4.8 Human study

Several OOC captions can appear over time for an image, each of them miscaptioning the image in a different way. We assess to which extent the output of AFC models constitutes a reusable artifact for humans to verify new captions about the same im-

	A	F1	κ
Group 1 - no AFC artifact	38.9	38.7	23.0
Group 1 - SNIFFER artifact	60.0	58.3	44.7
Group 2 - no AFC artifact	34.4	33.8	34.7
Group 2 - COVE artifact	85.6	83.0	60.8

Table 4: Human study results (%). κ is Fleiss’ κ . The best scores are marked in **bold**.

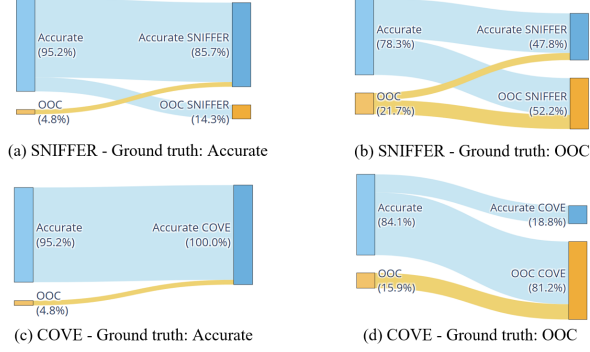


Figure 5: Change in veracity prediction before and after seeing SNIFFER (top) or COVE (bottom) artifacts, for accurate (left) and OOC (right) captions.

age. To the best of our knowledge, this is the first study of this important property of AFC artifacts. We compare two artifacts: the explanations generated by SNIFFER, which focuses on inconsistency detection, and the context predicted by COVE.

The study is split into two phases. (1) Given an image and an old OOC caption correctly detected by the AFC model, the human annotator has to classify three new captions about the same image. (2) The annotator classifies again the same new captions, this time given the AFC artifact generated for the old caption. We use the majority vote of the annotators in each phase as the label assigned to a new caption. Following Qi et al. (2024), we ask participants to provide confidence levels: not, somewhat, or highly confident.

We collect 30 OOC instances from the NewsCLippings test set and construct three new captions for each. This results in 22 and 68 accurate and OOC captions, respectively. Appendix I explains the creation of the new captions. Annotators instructions and examples are shown in Appendix J.

We recruited 6 students and provided the SNIFFER explanations to half of them and the COVE context to the others. Table 4 shows that both groups perform poorly without AFC artifacts. This means that knowing that the old caption is OOC does not inform the annotators much about the ve-

racity of the new ones. Hence, models that only provide a veracity score, like RED-DOT and AITR, have limited purpose beyond a specific (image-caption) pair. COVE contexts are more useful artifacts than SNIFFER explanations to verify new captions, as demonstrated by the larger increase in accuracy, Macro F1, and inter-annotator agreement (Fleiss’ κ (Fleiss, 1971)) in the second phase. We attribute this to COVE’s more comprehensive approach, predicting context items even if they are not relevant to verify the old caption, e.g., the caption does not contain a date, but COVE still predicts the true date. On the other hand, SNIFFER often explains the minimal set of inconsistent elements to detect an OOC caption, which might be insufficient to verify new captions.

Figure 5 shows the change in predictions between the two phases of the study. In the first phase, both groups label the majority of captions as accurate. Upon observing the COVE artifact, annotators successfully identify all accurate captions, and the number of detected OOC captions increases by more than 65 percentage points. By comparison, the SNIFFER artifact results in a 30.5 percentage points increase in detected OOC captions. Furthermore, observing the COVE artifact exclusively improves predictions by correcting errors from the first phase, whereas observing the SNIFFER artifact introduces new errors, shifting some correct predictions to incorrect ones.

This experiment confirms that COVE provides a reusable artifact to verify new captions, thanks to its comprehensive context prediction. We report aggregated confidence levels in Appendix K.

4.9 Error analysis

We manually analyze random samples of 200 instances from each test set and report the distribution of errors in Table 5. There are five error categories.

(1) Incorrect context items are predicted, which leads to veracity errors, in particular for accurate captions. In 5Pils-OOC, more than half of these errors stem from irrelevant Wikipedia entities detected by CLIP. This issue is especially pronounced in African and South Asian contexts, which are prevalent in 5Pils-OOC. In contrast, this problem does not arise in NewsCLippings, where most images have a Western context. Irrelevant or inaccurate web captions further contribute to context prediction errors.

(2) Context items are missing. This prevents Llama 3 from verifying all the atomic facts in the

	NewsCLiPPings	5Pils-OOC
Incorrect items	20.5	57.1
↔ Web captions error	6.8	14.3
↔ Wikipedia entities error	0.0	33.3
↔ Auto. captions error	9.1	7.1
↔ Knowledge gap error	4.6	2.4
Missing items	54.6	36.9
↔ Missing date	15.9	26.2
↔ Missing location	29.5	27.4
↔ Missing people	15.9	3.6
↔ Missing things	9.1	1.2
↔ Missing event	31.8	20.2
Rule-based error	2.3	0.0
Llama 3 veracity error	6.8	1.2
Error in ground truth	15.9	4.8

Table 5: Error distribution (%) on the test sets with few-shot Llama 3. ↔ indicates a sub-category.

caption. The most frequently missing items are the *date*, *location*, and *event*. While *people* is important in NewsCLiPPings, its absence has a smaller impact in 5Pils-OOC. This is because misrepresenting the individuals in the image is less frequent in real-world misinformation.

(3) Only one error in NewsCLiPPings is due to an instance passing the t_{match} veracity rule while actually being OOC. (4) In a few cases, the context items are correct and sufficient to predict the veracity, but Llama 3 makes a reasoning error. (5) We found some errors in the ground truth, where the caption is not a description of the image. Therefore, the instance is not suitable for veracity prediction.

This analysis highlights directions for future work, including a better similarity matching of Wikipedia entities for images with non-Western contexts and a better ranking of web captions based on their relevance. We provide three error examples in Appendix L.

5 Conclusion

We propose COVE, a new method to combat OOC misinformation that predicts first the true context of an image and leverages it to predict the veracity of its caption. COVE outperforms the SOTA on context prediction for all context items while being competitive with the best veracity models, even outperforming them on real-world data by up to 4.5 percentage points in Macro F1, showcasing the benefits of sequentially performing context and veracity prediction. Furthermore, our human study shows that the predicted context is a useful and reusable artifact for human users to verify new captions for the same image.

6 Limitations

We identify four limitations to this work.

(1) Similar to other QA methods in AFC (Pan et al., 2023b,a; Khaliq et al., 2024; Schlichtkrull et al., 2023), COVE requires several (M)LLM inference steps which are computationally expensive. However, the context predictions and automated captions can be generated in batches, improving inference speed. Furthermore, some steps of COVE can be removed depending on the computational budget available. For example, knowledge gap completion requires many LLM calls while being less critical to the pipeline than web caption and Wikipedia entity collection.

(2) We did not consider the use of closed-source LLMs like GPT4 (OpenAI, 2023) for context and veracity prediction. However, this work does not have the objective to provide a performance comparison between open and closed-source LLMs. COVE is LLM-agnostic, and we expect its performance to improve with LLMs that perform better on standard benchmarks and leaderboards.

(3) The context items of NewsCLiPPings and parts of the items of 5Pils-OOC are weakly labeled by decomposing accurate captions. However, there is no guarantee that the ground truth provides the most comprehensive summary of the context. In some cases, the context predictions might be correct and more detailed than the ground truth, e.g., predicting the correct location at the town level, while the ground truth is at the country level. The predictions are slightly penalized in the evaluation metrics when providing more precise answers despite them being correct. This limitation is also present in the 5Pils dataset (Tonglet et al., 2024).

(4) Some of the retrieved web captions may contain misinformation themselves or come from unreliable sources. However, there is no filtering mechanism implemented to remove web captions collected from unreliable websites. In Appendix M, we discuss a simple filtering mechanism that keeps web evidence from a manually defined list of reliable web domains, with a small negative impact on performance. Future work should consider the inclusion of models designed for evaluating the reliability of web evidence (Chrysidis et al., 2024) and their sources (Schlichtkrull, 2024).

7 Ethics statement

Intended uses COVE addresses the important societal problem of multimodal misinformation

by contextualizing images and detecting OOC captions. While AFC methods have made significant progress over the years, they are still prone to errors. In particular, the context predicted by COVE is still far from reaching sufficient quality, and as shown in Table 2, only three OOC captions out of four are detected on real-world data. Given the high negative impact of labeling misinformation as true information, and vice-versa, AFC methods like COVE should only be used as complementary assistants to a human FC expert.

Misuse potential COVE and other AFC methods show promising results in detecting misinformation. As a result, they could also be used in an adversarial setup by malicious actors to craft misinformation that is harder to detect by AFC methods and human fact-checkers. Nevertheless, we believe that the benefits of supporting the work of human fact-checkers with partial automation outweigh the risks caused by this malicious adversarial setup.

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A Object categories for automated captions

Table 6 provides the mapping from the object labels of the Google Vision API to our manually defined parent categories. Each category is associated with one captioning prompt that is provided to LlavaNext to caption the cropped image of the detected object.

B Definition of the context items

We consider seven context items which we define below. They are paired with a question and a set of relevant named entities, which are used to rank the web captions.

Source is the entity who created the image or first published it online (Tonglet et al., 2024). The corresponding question is “Who is the source of the image?”. The corresponding named entity is “ORG”.

Date is the time at which the image was captured or created (Tonglet et al., 2024). The corresponding question is “When was the image taken?”. The corresponding named entity is “DATE”.

Location is the place where the image was captured (Tonglet et al., 2024). The corresponding question is “Where was the image taken?”. The corresponding named entities are “FAC”, “GPE”, “LOC”.

Motivation is the reason why the source took the image, usually to report on a news event, although other intents are possible (Tonglet et al., 2024). The corresponding question is “Why was the image taken?”. The corresponding named entities are “EVENT”, “GPE”, “NORP”, “ORG”.

People is the people that can be seen in the image. The corresponding question is “Who is shown in the image?”. The corresponding named entity is “PERSON”.

Things is a broad category that includes every entity that is shown in the image and not a human being. The corresponding question is “Which animals, plants, buildings, or objects are shown in the image?”. The corresponding named entities are “FAC”, “LOC”, “PRODUCT”.

Event describes the circumstances surrounding the image. The corresponding question is “Which event is depicted in the image?”. The corresponding named entities are “EVENT”, “NORP”.

Dufour et al. (2024) showed that the context items that are the most frequently altered when creating OOC captions are the *date* and the *event*, around 25% each, followed by *Location* around 18%. *People* and *things* reach together around 15%.

C Examples of knowledge gap completion

Figure 6 shows two examples of knowledge gap completion on 5Pils-OOC. The question generation and the retrieval of a relevant Wikipedia passage allows to predict the location in the first image and the date in the second one.

D Prompts for caption decomposition

To obtain the ground truth context items for NewsCLippings and for parts of 5Pils-OOC, we task Llama 3 to decompose the accurate caption of the image as a dictionary with context items as keys. Figure 7 shows the prompt.

E Creation of 5Pils-OOC

5Pils (Tonglet et al., 2024) is the first real-world misinformation dataset that provides labels for context items. The ground truth context of the image is obtained by extracting the context items from an FC article written by human experts. To use

Category	Objects	Prompt
general	entire image	Answer in one to three sentences: what are the people, objects, animals, events, texts shown in the image?
people	Person	Who is shown in the image?
animals	Animal, Bird, Cat, Dog, Fish	Which { } species is shown in this image?
buildings	Building, Stadium, Bridge, Castle	Which { } is shown in this image? Provide a location if possible.
flags	Flag	Which flag is shown in this image?
food	Food, Drink, Fruit	Which { } is shown in this image?
sports	Basketball, Baseball bat	What are the teams playing in this game?
	Baseball glove, Football, Rugby ball	
transports	Airplane, Boat, Bus, Car, Helicopter	Which { } model is shown in this image?
	Motorcycle, Ship, Tank, Train, Truck, Van	
weapons	Weapon	Which weapon model is shown in this image?

Table 6: Parent categories of detected objects with their captioning prompts.

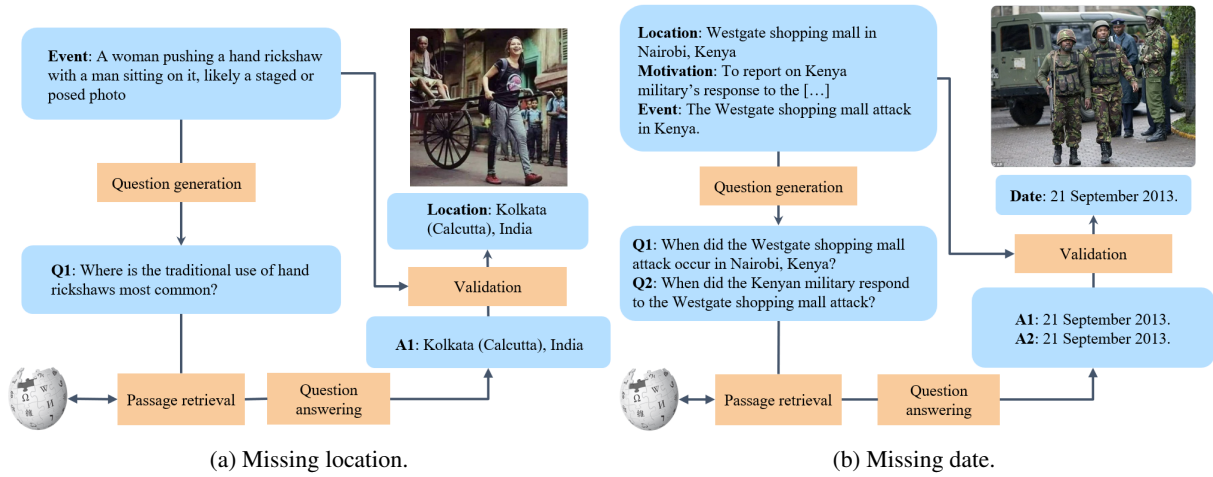


Figure 6: Two examples of successful knowledge gap completion on 5Pils-OOC.

Caption decomposition prompt

You are a helpful assistant who provides a structured summary of the content of an image. The summary should take the form of a JSON file with the following key entries: “people” (the people shown in the image as a list), “object” (the objects shown in the image as a list), “event” (the event depicted in the image), “date” (the date when the image was taken), “location” (the location where the image was taken), “motivation” (the reason why the photographer took the image, such as for documenting, reporting, or for personal or organizational purposes. This reflects the photographer’s intent. Do not assume it if not specified), “source” (the original author of the image, a person or an organization name).

You need to provide the answer to each of those items based on the caption of the image. Note that the caption does not always describe the image content. If there is no information in the caption for one of the key entries, you should write “not enough information” for that entry of the JSON file. Your answer should only contain the JSON file as a dictionary.

Figure 7: Prompt template for caption decomposition with Llama 3.

	Source		Date		Location				Motivation		
	(RL)	(M)	(EM)	(Δ)	(RL)	(M)	(CO Δ)	(HL Δ)	(RL)	(M)	(BertS)
NewsCLippings											
5Pils baseline	5.0	3.4	23.1	28.9	34.4	32.7	38.7	33.1	1.0	1.5	9.6
COVE	12.2	8.1	30.7	41.5	44.2	42.4	51.6	43.1	5.6	11.6	65.3
5Pils-OOC											
5Pils baseline	0.4	0.3	0.5	1.8	15.2	12.1	21.8	16.8	3.0	3.0	62.0
COVE	0.9	0.6	1.1	7.0	18.6	16.7	28.9	22.5	17.1	15.1	56.1
	People			Things				Event			
	(R)	(P)	(F1)	(RL)	(M)	(BertS)	(RL)	(M)	(BertS)		
NewsCLippings											
5Pils baseline	38.9	42.6	39.9	12.2	9.9	62.6	14.1	15.4	43.6		
COVE	48.2	51.5	49.0	9.7	10.9	63.0	14.6	22.4	68.1		
5Pils-OOC											
5Pils baseline	12.1	14.1	12.8	6.9	4.9	63.9	5.3	4.8	65.0		
COVE	20.8	21.5	20.5	6.8	7.2	61.2	7.0	9.4	51.7		

Table 7: Detailed context prediction results on the test sets (%). Best results are marked in **bold**.

Accurate caption generation prompt

You are given a date, a location, and a motivation describing an image. Combine the 3 in one sentence of maximum 30 words. Write the facts only, avoid journalistic style and adjectives, avoid introducing new information.
Date : { } , Location : { } , Motivation : { }

Figure 8: Prompt to generate an accurate caption with GPT4, given the ground truth context items.



Figure 9: A composite image, where the sub-images have distinct true contexts.

5Pils for context and veracity prediction, we need to make some adjustments.

First, the scope of 5Pils is broader than ours. It includes other types of misinformation than OOC images, namely, manipulated and fake images. Hence, we start by removing all images that are manipulated or fake using the “type of image” metadata field. Furthermore, 5Pils does not provide accurate captions. We generate accurate captions with GPT4 by combining the ground truth context items. This requires at least the *motivation* item and one or two out of *date* and *location*. Images that do not satisfy those label criteria are discarded. Figure 8 shows the prompt to generate accurate captions. Finally, some images in 5Pils are composite, that is, they are a collage of more than one image next to each other. An example is shown in Figure 9. Such images are considered out-of-scope.

F Detailed context prediction evaluation

Table 7 complements the results of Table 1 by showing additional metrics for the context items defined in Tonglet et al. (2024). This means computing the RougeL (RL) score (Lin, 2004) for *source*, the exact match (EM) for *date*, the RougeL and Meteor scores for *location*, and the RougeL and BertScore (BertS) (Zhang et al., 2020) for *motivation*. Furthermore, we compute HL Δ for *location*, which is inversely proportional to the hierarchical distance in the GeoNames ontology between the prediction and ground truth. We also compute additional metrics for the new items introduced in this work. For

Context prediction prompt

You are a helpful assistant who answers questions about an image based on captions from the web, automatically generated captions, and visual entities.

You are given captions from web pages that use the image.

You are also given two automatically generated captions. The first caption provides a global description of the image, sometimes including people but without naming them.

The second caption provides more details about the people in the image, including their names and short biographies when that information is available.

You are also given visual entities which may be relevant to the image without certitude.

Here are a few examples:

[Demonstrations]

Web captions that may be relevant: {}

Caption 1 - Global description: {}

Caption 2 - More details about the people in the image: {}

Visual entities that may be relevant, without certitude: {}

Question: {}

Answer in one sentence. Answer with one word ("unknown") if the information is not provided.

Answer: {}

Figure 10: Prompt template for context prediction with Llama 3.

people, we compute the Recall (R) and Precision (P). For *things* and *event*, we add the RougeL and Bert scores. Consistent with our observations in Table 1, we observe that COVE outperforms the baseline (Tonglet et al., 2024) for most metrics.

G Hyperparameters and model versions

Web captions collection We set t_{match} to 0.92 and t_{non_match} to 0.7 for web images. Visual entities are included if their score is at least 0.1.

Wikipedia entities collection The following types of named entities are extracted from the caption: “PERSON”, “FAC”, “PRODUCT”. We set k , the number of nearest entities to retrieve from the OVEN index (Hu et al., 2023a) to 5. We set t_{wiki_text} to 0.23. If the entity is a “PERSON”, t_{wiki_image} is set to 0.92. Otherwise, it is set to 0.7.

Automated captions generation We keep *Person* objects detected with confidence scores above 0.8.

Context prediction We set l , the number of web captions to provide as input, to 10.

Knowledge gap completion For each question, we retrieve one Wikipedia passage, if its relevance score (Semnani et al., 2023) is above 20.

Model and versions We use the HuggingFace’s transformers (Wolf et al., 2020) and vLLM (Kwon et al., 2023) libraries to load, train, and make inferences with models. Named entities are detected using the Spacy model *en_web_core_lg*. We create the OVEN index with the FAISS library (Douce et al., 2024). We use the CLIP-ViT-L14 (*openai/clip-vit-large-patch14*) version of CLIP. We also considered CLIP-ViT-B32 (*openai/clip-vit-base-patch32*) and SIGLIP (*google/siglip-so400m-patch14-384*) (Zhai et al., 2023). On the validation set of NewsCLippings, both SIGLIP and CLIP-ViT-L14 achieve the same accuracy, but CLIP-ViT-L14 has a slightly higher R_{OOC} . Accuracy with CLIP-ViT-B32 is 1 percentage point lower. To compute similarities between the image and Wikipedia images for “PERSON” entities, we use the face-net library.⁶ For COVE, SNIFFER, and the baseline of Tonglet et al. (2024), we use the *meta-llama/Meta-Llama-3-8B-Instruct*, *llava-hf/llava-v1.6-mistral-7b-hf*, and *microsoft/deberta-v3-large* versions of Llama 3, LlavaNext, and DebertaV3, respectively. We set the temperatures to 0 for reproducibility. We fine-tune DebertaV3 for 5 epochs, using a batch

⁶pypi.org/project/face-recognition/

Question generation prompt

You are a helpful assistant. You are given context information about an image. The date of the image is unknown.

Your task is to generate one to three world knowledge questions that will help determine when the image was taken. The questions should either start with “When” or “On which date”. The question should be self-contained and specific enough to be answerable based on world knowledge.

If no self-contained and specific question can be generated, your response should be “No questions can be generated given the context”.

You are provided with examples below.

[Demonstrations]

Context information: { }

Generated questions (up to 3): { }

Figure 11: Prompt template for knowledge gap completion - question generation with Llama 3.

Question answering prompt

You are a helpful assistant. You are given a question that requires world knowledge to be answered. Your task is to provide a specific answer to the question in 1 or 2 sentences based on available knowledge from Wikipedia. Your answer should be a date at the day, month or year level.

If the question cannot be answered based on the available knowledge, your response should be “Unknown”.

You are provided with examples below.

[Demonstrations]

Wikipedia knowledge: { }

Question: { }

Answer: { }

Figure 12: Prompt template for knowledge gap completion - question answering with Llama 3.

Validation prompt

You are a helpful assistant. You are given context information about an image, as well as world knowledge information.

Your task is to estimate when the image was taken or provide a plausible time range using the context and world knowledge information.

If the date cannot be derived from the context and world knowledge, your response should be “Unknown”.

You are provided with examples below.

[Demonstrations]

Context: {}

World knowledge: {}

When was the image taken?

Answer: {}

Figure 13: Prompt template for knowledge gap completion - validation with Llama 3.

Veracity prediction prompt

You are a helpful assistant that verifies if a caption is accurate for an image or if it is an attempt at out-of-context misinformation. Instead of the image itself, you are given structured context information about the image. You need to verify whether the caption is supported by the context information.

Here are a few examples:

[Demonstrations]

Context information: {}

Caption to verify: {}

Given the context information, is the caption accurate or is it out-of-context?

If there are too many unknown elements to provide a clear decision, you can answer that the accuracy of the caption is “unknown”, potentially leaning more towards accurate or out-of-context.

Provide a detailed reasoning. Then provide your answer strictly among the following choices: “accurate”, “unknown, probably accurate”, “unknown”, “unknown, probably out-of-context”, “out-of-context”.

Reasoning: {}

Answer: {}

Figure 14: Prompt template for veracity prediction with Llama 3.

size of 4, a weight decay of 0.01, and a learning rate of 5e-6. All experiments are conducted with one A100 GPU.

H COVE - Llama 3 prompt templates

Figure 10 to 14 show the prompts used for context prediction, knowledge gap completion, and veracity prediction with Llama 3. Each prompt starts with a task description, followed by a set of 4 to 8 demonstrations. For veracity predictions, there are two sets of demonstrations, one for instances with web captions and one for those without web captions. The template shows the structure of the demonstrations and the test instance. {} are replaced by the value of the demonstrations or the test instance. For knowledge gap completion, we provide the prompts used for *date* prediction.

I Human study corpus creation

The human study corpus contains 30 images with OOC captions sampled from the NewsCLippings test set. We restrict the selection to images with at least six predicted context items. For each image, we create three new captions. This results in a corpus of 90 new captions, of which 22 are accurate and 68 OOC. The number of accurate captions per image ranges from 0 to 2.

Accurate captions are either the original caption from Visual News (Liu et al., 2021) or a hand-written paraphrase based on additional context from the source news article.

Parts of the OOC captions are sampled from other splits of NewsCLippings that contain the same image. Others are written by hand by paraphrasing the original caption from Visual News and altering key context items based on the source news article, following the OOC misinformation techniques detailed in Dufour et al. (2024).

The low accuracy of the annotators in the absence of AFC artifacts and the moderate increase given the artifacts confirm the challenging nature of the created corpus.

J Human study instructions and examples

The following instructions were provided to the participants in the first phase: “Your task is to decide if a caption is accurate for an image or if it is out-of-context. Out-of-context means that the caption is (partially or totally) not matching the image (in terms of people, event, date, location,

	Not confident	Somewhat confident	Highly confident
Group 1 - no AFC artifact	45.9	37.8	16.3
Group 1 - SNIFFER artifact	14.8	54.4	30.7
Group 2 - no AFC artifact	38.5	48.9	12.6
Group 2 - COVE artifact	8.1	33.0	58.9

Table 8: Confidence levels of the annotators (%) in different setups. Each row sums to 100%.

objects, ...). The caption may or may not describe a real event, but it is not accurate for the given image. You will be given an image and a caption that has already been assessed as out-of-context, based on that you need to classify 3 new captions as Accurate or out-of-context, and indicate your confidence level. Important : do not search for information online or conduct reverse image search with the image. You are expected to answer based only on the image and the previously fact-checked caption.”

Afterward, the participants are given the instructions for the second phase. If they are given the COVE artifact, they receive the following instructions: “The task is the same, and you will see the same images and captions to verify but this time, you are also given a summary of the image context, generated with the AFC method COVE. You can now update your verdict and confidence scores based on this additional input. ”. If they are given the SNIFFER artifact, they receive these instructions: “The task is the same, and you will see the same images and captions to verify but this time, you are also given the explanations generated with the AFC method SNIFFER. SNIFFER explanations consist of 2 parts: Internal checking: the model compared the image with the previously fact-checked caption, as well as a set of relevant visual entities, and detected inconsistencies if any. External checking: the model compared the previously fact-checked caption with relevant web evidence found by doing a reverse image search. You can now update your verdict and confidence scores based on this additional input.”

Figure 15 shows two examples of the corpus.

K Human study confidence levels

We report in Table 8 the non-aggregated confidence levels of the participants. For both groups, the “Highly confident” level is not frequent during the first phase. After seeing the artifact, the most frequent level becomes “Somewhat confident” for SNIFFER and “highly confident” for COVE.

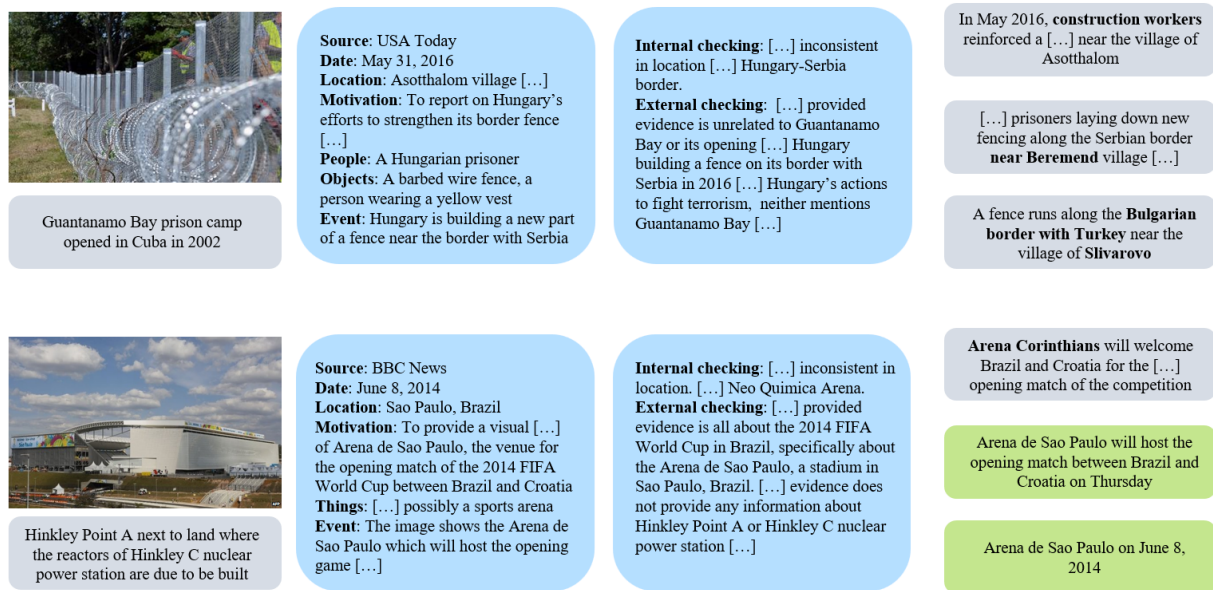


Figure 15: Human study examples. From left to right, the image with the old caption, the COVE artifact, the SNIFFER artifact, the three new captions. **Accurate** captions are shown in green, **OOO** captions in grey.



Figure 16: Error examples with Llama 3. Each row shows the image (left), the context prediction (center), the caption (upper right), and the veracity prediction (lower right). **Accurate** captions are shown in green, **OOO** captions in grey.

L Error examples with COVE - Llama 3

Figure 16 provides three error examples of Llama 3 on the NewsCLIPPings test set. In the first example, the predicted context is missing the *date* and *location* items. Llama 3 wrongly predicts the caption as OOC, given that not all atomic facts are supported by the context. The second image is an example of context prediction error propagating to veracity prediction. While Edisto Island is the correct location of the image, the predicted *location* is incorrect. As a result, the caption is wrongly predicted as OOC. The third example is the opposite of the first example. The predicted context is not sufficient to verify all atomic facts, but the caption is wrongly predicted as accurate.

M Filtering mechanism experiment

We conduct an experiment on 5Pils-OOC where we only use web evidence that belongs to a manually curated list of trustworthy sources. The sources are: theguardian.com, usatoday.com, nytimes.com, washingtonpost.com, reuters.com, indiatimes.com, bbc.com, cnn.com, nbcnews.com, thetimes.co.uk, and apnews.com. By selecting web evidence from these sources only, the accuracy decreases by 2.3 and the Macro F1-score by 2.8 percentage points. Only R_{OOC} decreases while R_{ACC} remains unchanged. Despite the simplicity of this filtering approach, the decrease in veracity prediction performance is relatively small, indicating that a more advanced filtering mechanism could provide results equivalent to an approach without filtering or even outperform it.